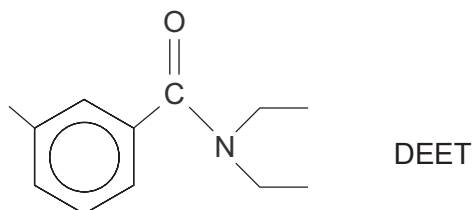


SECTION B

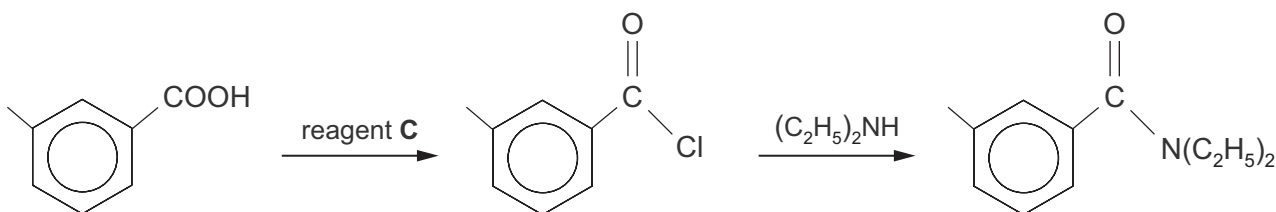
Answer **all** questions in the spaces provided.

9. Insect repellents are materials that can be applied to the skin or to clothing in order to provide protection against biting insects.

(a) N,N-Diethyl-3-methylbenzamide, commonly known as DEET, has proved to be one of the most effective insect repellents.



- (i) One method of preparing DEET is by the following reaction sequence.



Identify reagent **C** that can be used to convert 3-methylbenzoic acid to 3-methylbenzoyl chloride. [1]

- (ii) DEET is a colourless liquid in white light.

Explain, in terms of the electromagnetic spectrum, why DEET is colourless. [1]

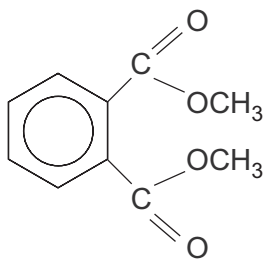
- (iii) The mass spectrum of DEET shows a molecular ion at m/z 191. It also shows a prominent peak at m/z 91.

Suggest a formula for the fragment that is **lost** so that the remaining fragment has m/z 91. Show your working. [2]

Formula



- (b) An alternative insect repellent is dimethyl benzene-1,2-dicarboxylate, DMP.



DMP

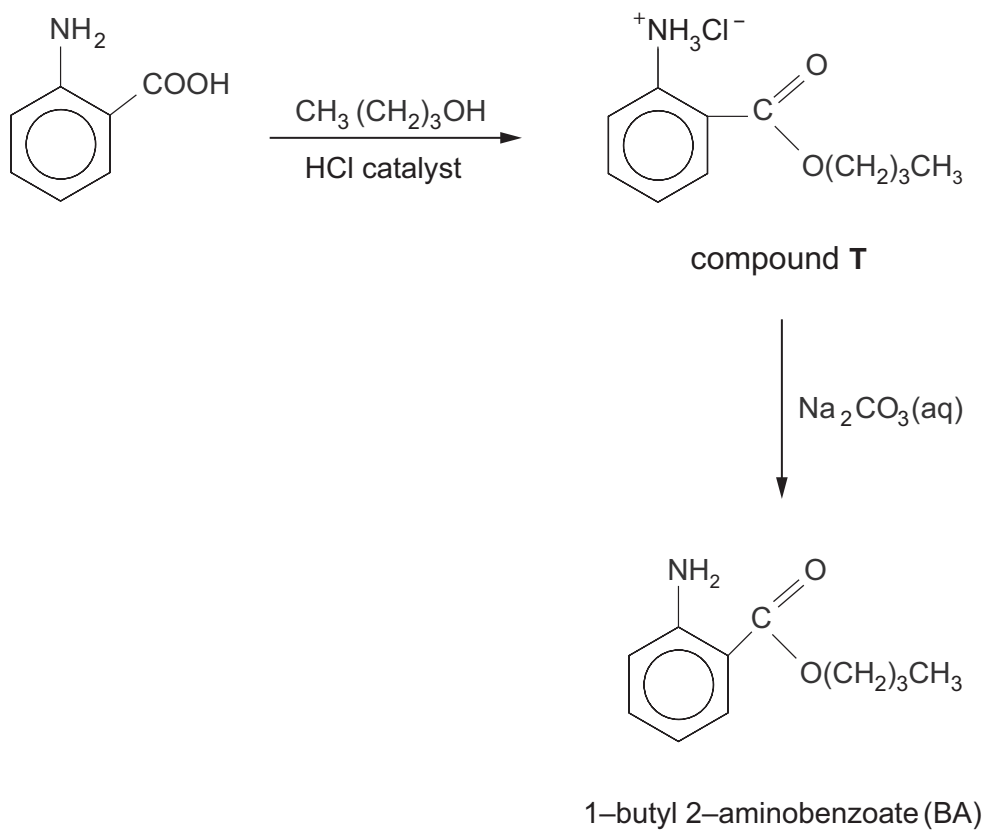
This can be made by the oxidation of 1,2-dimethylbenzene, followed by acidification of the resulting product and then esterification.

- (i) State the oxidising agent used. [1]

- (ii) Explain why it is necessary to acidify the product of this oxidation. [1]



- (c) Another insect repellent is the 1-butyl ester of 2-aminobenzoic acid, known as BA. This can be produced from 2-aminobenzoic acid by esterification with butan-1-ol.



- (i) Draw the zwitterion structure of 2-aminobenzoic acid.

[1]

- (ii) Explain why compound **T** is initially produced in this reaction.

[1]

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- (iii) The ester BA is obtained from compound T by reacting it with sodium carbonate solution. It is then removed from the mixture by extracting it with the solvent ethoxyethane. After drying the ethoxyethane extract, the solvent is removed by distillation. Ethoxyethane has a boiling temperature of 35 °C and is very flammable.

I. Suggest the origin of the water removed during the drying process. [1]

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II. Suggest how the ethoxyethane solution of BA could be safely heated to remove the solvent. [1]

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- (d) There is considerable interest in 'green' methods for the production of organic compounds. Recent studies have shown that BA can be produced from 2-aminobenzoic acid and butan-1-ol at 37 °C, using an enzyme as a catalyst.

- (i) In a small scale experiment 4.0×10^{-4} mol of 2-aminobenzoic acid gave a 5 % yield of BA.

Calculate the number of moles of BA that were produced. [1]

$n(\text{BA}) = \dots\dots\dots$ mol

- (ii) The esterification of 2-aminobenzoic acid is a reversible reaction and eventually the mixture will reach equilibrium. Use the information below to suggest why the addition of hexane to the stirred aqueous mixture of reactants increases the yield of the product BA. [3]

- hexane and the aqueous reaction mixture are immiscible
- 2-aminobenzoic acid is more soluble in water than in hexane
- BA is more soluble in hexane than in water

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